

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Pharm Examination

April-May 2015

PH410 Medical Chemistry – IV

Date: 27.04.2015 , Monday

Time:10:00 a.m to 01:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the following

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|---|-----|---|----|
| A | i. | Give the classification of sulphonamides with their structures. Write a synthetic pathway of Sulphamethoxazole. | 05 |
| | ii. | Explain Structural Activity Relationship (SAR) & Mechanism of action of Sulphonamides. | 05 |
| B | i. | Write a mechanism and structure of 1 st line Antimycobacterial agents. | 05 |
| | ii. | Write a chemical classification of Antifungal agents. | 05 |
| C | i. | Write a detail note on Disinfectants. | 05 |
| | ii. | Give the chemistry of Immunosuppressive agents. | 05 |

Q 2 Attempt any **TWO** of the following

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|---|-----|---|----|
| A | | Give the classification of Antineoplastic agents with structures. Explain mechanism of action and SAR Alkylating agents. | 10 |
| B | i. | Explain SAR of Quinolone derivatives. | 05 |
| | ii. | Classify Anti-malarial agent with atleast one structure from each class. | 05 |
| C | i. | Classify Antiviral agents with their structure. Enlist Nucleoside Reverse Transcriptase Inhibitors (NRTIs) used in treatment of HIV with structure. | 05 |
| | ii. | Classify Anti Amoebic Agent with structure of each class. | 05 |

SECTION- II

Q 3 Attempt any **TWO** of the following

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|---|-----|--|----|
| A | i. | Write a mechanism of action, metabolism pathway and synthesis of Chloramphenicol. | 05 |
| | ii. | Explain SAR of Chloramphenicol in details. | 05 |
| B | i. | Write a detail note on Quantitative Structure Pharmacokinetic Relationship (QSPR). | 05 |
| | ii. | Write a brief note on bioinformatics. | 05 |
| C | i. | Explain SAR of Penicillin derivatives in details. | 05 |
| | ii. | Write a chemistry of Macrolide antibiotics. | 05 |

Q 4 Attempt any **TWO** of the following

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|---|-----|--|----|
| A | i. | Explain SAR of Tetracycline antibiotics. | 05 |
| | ii. | Write a detail note on purification technique of combinatorial library. | 05 |
| B | i. | Write a chemistry of Aminoglycosides in details. | 05 |
| | ii. | Give the classification and mechanism of Polyene antibiotics. | 05 |
| C | i. | Explain principle and importance of combinatorial chemistry in new drug discovery. | 05 |
| | ii. | Enlist various approaches of combinatorial library synthesis. Explain parallel synthesis in details. | 05 |